TRACE TABLE FOR AMICABLE PAIR (220, 284).

Date:

Checking the number 220.

As given in the question, we can use square root trick, we can add divisors upto square root: $\sqrt{220} = 14$, so we will check from i=1 to 14.
so, each time sum will be increased by i + 220/i.

i	does i divide 220	divisors	running sum
1	Yes.	1	1
2	Yes	2, 110	113
3	N	-	113
4	Yes	4, 55	172
5	Yes	5, 44	221
6	N	-	221
7	N	-	221
8	N	-	221
9	N	-	221.
10	Yes	10, 22	253
11	Yes	11, 20	284
12	N	-	284
13	N	-	284
14	N	-	284

Therefore, $s(220) = 284$.

Date:

Checking 284. $= \sqrt{284} = 16.$ $(i = 1 \text{ to } 16)$.

i	does i divide 284	divisors	sum.
1	Yes	1	1
2	Yes	2, 142	145
3	No	–	145
4	Yes	4, 71	~~145~~ 220
5-16	No	–	stop!

$s(284) = 220.$

conclusion:

220, 284 are amicable pairs!

→ $s(220) = 284$

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